

Daphne: Divergence-Aware Federated Learning for Task-Adaptable Health Analytics

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Wearable consumer electronics generate a substantial amount of health data. However, much of this data remains underutilized for deriving meaningful health insights due to its distributed nature, privacy concerns, and lack of labels. Federated Learning (FL) offers a promising solution to effectively mine such health-related data. However, conventional FL methods often fail due to divergence caused by the inherent heterogeneity in health analytics data collected from numerous client devices. To address these challenges, we propose Daphne, a novel FL framework that provides comprehensive solutions for divergence at three levels. First, Daphne introduces client-contrastive learning to manage data heterogeneity at the *representation level*. Additionally, Daphne employs unsupervised curriculum learning to reconcile divergence at the local *training level*. Finally, Daphne introduces a learnable weighted aggregation to mitigate residual divergence at the global *aggregation level*. Extensive experiments demonstrate that Daphne learns robust health data representations, adaptable for tasks such as emotion recognition, sleep quality monitoring, and fatigue detection, outperforming conventional FL and supervised approaches.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computing methodologies** → **Distributed computing methodologies**.

Additional Key Words and Phrases: Federated learning, health analytics, multi-modal sensing, model divergence.

ACM Reference Format:

Yuxuan Weng, Jinyang Huang, Menglan Hu, and Tianyue Zheng[✉]. 2018. Daphne: Divergence-Aware Federated Learning for Task-Adaptable Health Analytics. *ACM Trans. Sensor Netw.* 37, 4, Article 111 (August 2018), 24 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

With rapid advancements in hardware and software, increasingly compact and powerful consumer electronics, such as smartphones [3, 7, 19, 22–24, 60], watches [2, 52], glasses [26, 27, 74], and gloves [41], are becoming ubiquitous. These devices continuously monitor a range of physiological signals, including photoplethysmogram (PPG), blood oxygen saturation, electrodermal activity (EDA), and electromyography (EMG). The widespread adoption of continuous and multimodal monitoring generates substantial volumes of data. For instance, wearable monitors alone generate an estimated 335 petabytes of medical data every month, according to Statista [56]. Despite the potential richness of these datasets, they remain underutilized due to several critical challenges. First, the decentralized nature of these devices

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means that data often remains siloed on local devices, hindering comprehensive analysis. Second, privacy concerns preclude sharing of sensitive health data across devices. Third, annotation scarcity further diminishes the utility of collected data.

The collected data can provide valuable insights into both the physiological and psychological conditions of the users [8, 21, 25, 59, 65]. For instance, photoplethysmography (PPG) signals can detect cardiovascular issues [13, 50, 73], while electrodermal activity (EDA) sensors monitor central nervous system activity by measuring sweat gland responses [1, 16, 66]. Additionally, electromyography (EMG) sensors measure muscle activation and mechanical load [37, 40]. By integrating information from these diverse signals, it is possible to drive health applications such as cardiovascular diagnosis [28, 50], emotion detection [46, 67], sleep monitoring [9, 20, 61], and fatigue detection [4, 43, 48]. Furthermore, these collected signals can be processed through representation learning [36, 38, 44, 55], allowing for the creation of a unified representation that enhances the development of a **task-adaptable** system. Such a system can be versatile and capable of servicing a wider range of tasks.

Therefore, it would be a huge advancement if one can mine useful health insights from the underutilized health data generated by consumer electronics [10, 14, 35, 49, 57, 63, 68, 75]. Federated learning (FL) [15, 31, 39, 45, 51, 62, 64, 70, 72, 76] offers a promising solution to fully leverage the distributed, privacy-sensitive data for health analytics. In this emerging distributed machine learning paradigm, a central server coordinates multiple client devices that share only local model parameters. By avoiding the transmission of locally collected physiological signals, FL achieves both privacy preservation and communication efficiency during training. However, existing FL systems designed for health analytics are often highly task-specific. For example, ADMarker [45] uses multi-modal data from ambient in-home sensors in an end-to-end system to perform the specific task of detecting interpretable digital biomarkers for subsequent Alzheimer’s disease analysis; and FedTherapist [53] leverages linguistic expressions from a user’s smartphone for the singular purpose of predicting mental health states. An extension of FL, federated representation learning (FRL), further addresses annotation scarcity issue by combining FL with semi/un-supervised learning to learn a unified representation, readily to be for various downstream tasks.

Challenges. Although several FRL proposals exist [58, 71], they either combine FL with unsupervised learning [58] in a straightforward manner, or only address the basic class misalignment issue [71], without adequately processing the highly heterogeneous real-life health data. Specifically, these proposals lack a systematic approach to handle heterogeneity-induced divergence across different FL levels. At the *representation level*, covariate shifts [5] in the raw data, such as heart rate discrepancies among subjects, lead to misaligned representations and further diverge clients towards different local optima. At the *local training level*, representation learning without the constraints of annotations can result in unstable optimization trajectories on a non-smooth loss landscape. Finally, at the *global aggregation level*, abnormal device operations (e.g., incorrect/idle wearing, sensor malfunctions) may produce low-quality or invalid data. Without proper aggregation techniques to filter out such data, the learning process experiences further divergence.

Our solutions. To address these challenges, we present Daphne, a divergence-aware FRL framework for health analytics. Daphne learns robust representations from the decentralized data while progressively preventing divergence throughout different FRL levels, as depicted in Fig. 1. First, Daphne mitigates heterogeneity-induced divergence at the representation level. It employs client-contrastive learning, which compares latent spaces across various clients, addressing covariate shifts in the health data and ensuring consistent representations across the clients. Second, Daphne further reconcile divergence by stabilizing local optimization trajectories of the clients using unsupervised curriculum learning. By gradually increasing training complexity through adjustable mask lengths in the masked autoencoder, Daphne helps the clients navigate on the non-smooth loss landscape in a more stable manner. Finally, Daphne addresses

the remaining divergence with a learnable weighted mechanism for global aggregation. This mechanism adjusts weights based on client data quality, reducing the impact of low-quality data clients and promoting global convergence. Together, these three strategies enhance convergence throughout the FRL process, enabling the effective use of unlabeled data from the wearable devices. Our key contributions can be summarized as follows:

- To the best of our knowledge, Daphne is the first FRL system that systematically addresses the divergence issue to achieve task-adaptable health analytics.
- We develop a client-contrastive learning method that mitigates the impact of covariate shifts in the data, promoting globally consistent representations.
- We design an unsupervised curriculum learning approach that incrementally increases learning difficulty, ensuring stable optimization trajectories during local training.
- We introduce a learnable weighted mechanism for global aggregation, thereby reducing the divergence caused by abnormal client data.
- We evaluate Daphne through extensive experiments. Promising results confirm Daphne’s ability to enhance the convergence of FL and enable various health-related tasks.

In the following, § 2 introduces the background and motivation of Daphne. § 3 presents the system design of Daphne. § 4 introduces the system implementation and datasets used by Daphne. § 5 reports the evaluation setup and results. Finally, § 6 concludes the paper with future directions.

2 BACKGROUND AND MOTIVATIONS

In this section, we explore the heterogeneity issues at the *representation*, local *training*, and global *aggregation* levels that may lead to client divergence and performance degradation in FRL training to motivate the design of Daphne.

2.1 Data Covariate Shift

The health-related data collected by wearable sensors can exhibit varying features among different individuals, even if they are in identical physiological and psychological states. This variability is attributed to the unique characteristics of each subject. Technically, these type of data heterogeneity is termed covariate shift, where the feature distribution $P(X)$ varies across different clients, yet the conditional distribution of the labels Y given the features X , denoted as $P(Y|X)$, remains consistent across all clients. For instance, in emotion analysis using PPG signals, a standard technique involves

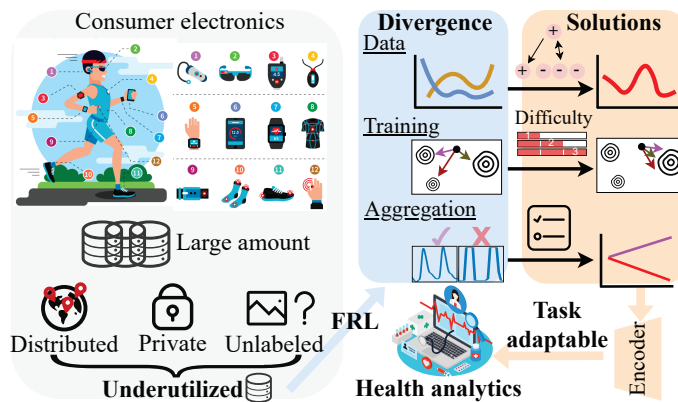


Fig. 1. Overview of Daphne.

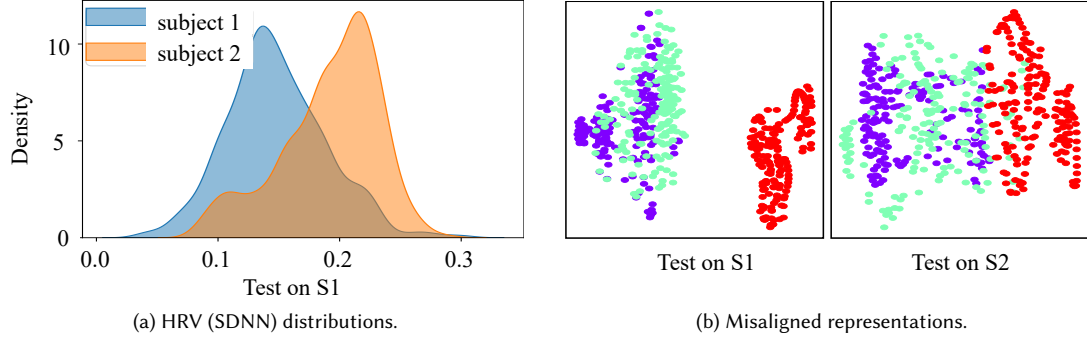


Fig. 2. Data covariate shift.

deriving heart rate variability (HRV) from the raw data. Typically, higher HRV indicates positive emotions, while lower HRV suggests negative emotions. However, HRV distributions can differ significantly between subjects. For example, Fig. 2a shows continuous 24-hour HRV monitoring of two subjects experiencing various emotions. The data reveals that subject 1 has a lower average HRV, whereas subject 2 exhibits a higher average HRV and a skew towards higher values. This disparity implies that models trained on heterogeneous data may learn subject-specific HRV thresholds, limiting their generalizability to new individuals and hindering the effectiveness of FRL.

To further illustrate the impact of covariate shift on FRL, we begin by training an encoder using the training data from subject 1. Subsequently, we process the test data from both subject 1 and 2 through this encoder. Two dimensions of the resulting representations are then visualized using t-SNE, as depicted in Fig. 2b. The visualization reveals a stark contrast between the two sets of test data. For test data that originates from the same distribution as the training data (subject 1), the resulting representations form well-defined clusters with distinct boundaries. However, for test data from a different distribution (subject 2), these cluster boundaries become blurred and cannot be aligned with that from subject 1. This observation clearly indicates that covariate shift in the original data results in misalignment at the *representation level*, consequently leading to divergence among the clients in FRL. To mitigate such divergence, it is imperative to develop a method that forces the model to automatically extract representations that remain consistent across the clients' data distributions.

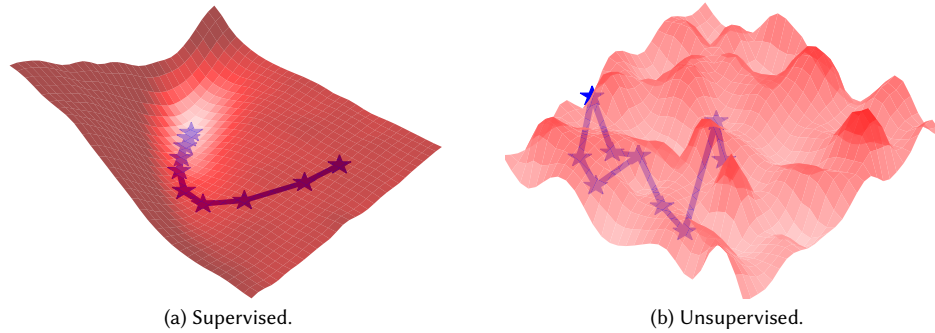


Fig. 3. Loss landscape and optimization trajectory.

2.2 Instability of Unsupervised Learning

Given that Daphne is a task-adaptable system, it faces challenges when deployed on wearable devices. Users, as non-experts, often lack the necessary knowledge and motivation to perform health-related annotations when encountering multiple tasks. Unsupervised representation learning [69] emerges as a promising solution. Without supervisory signals, it can discover useful representations that are readily adaptable to different tasks through fine-tuning. Specifically, employing a masked autoencoder (MAE) with a high fixed masking proportion (e.g., 50% or 75%) turns out to help learn representations with strong generalization capabilities [18]. However, due to the absence of supervisory constraints, unsupervised representation learning may become unstable, potentially falling into local minima during optimization.

To illustrate the instability of unsupervised representation learning on health-related data, we compare the loss landscapes and optimization trajectories between supervised (using annotations for emotion recognition) and unsupervised learning in Fig. 3. We utilize the random direction and filter-wise normalization method [32] for visualization. One may readily observe that in supervised learning, the loss landscape is smooth, leading to efficient optimization. In contrast, unsupervised representation learning faces numerous local minima due to multiple potential but viable data representations, resulting in a chaotic loss landscape and causing the optimization trajectory to fluctuate and navigate inefficiently. When the unsupervised representation learning is combined with FL, variations in data distribution across clients exacerbate these fluctuations, leading to overall optimization divergence. This issue urges us to design a method at the local *training level* to stabilize the optimization trajectories of representation learning.

2.3 Abnormal Device Operations

One fundamental assumption of FRL is the validity of the collected data. However, this assumption can be compromised by abnormal device operations during the use of wearable sensors. Incorrect wearing and sensor malfunctions are two common issues that can undermine data integrity. For example, during intense physical activity, the attachment of a sensor may loosen, preventing it from maintaining stable contact with the human body and leading to significant signal variations. Additionally, sensor malfunctions, such as damaged ADC or amplifiers, can result in signal distortion, clipping, and oversaturation. As illustrated in Fig. 4a, data collected under these abnormal conditions exhibit numerous artifacts and cannot provide useful health-related information, thus failing in facilitating the training of a FRL model. Unfortunately, it is not practical to identify and exclude these erroneous signals at the client-side due to resource constraints and privacy concerns [17, 42, 47].

Fortunately, this divergence in data quality can be identified at the model aggregation level. To illustrate this, we train 10 models using normal data and another 10 models using abnormal data. We then reduce the model parameters to two dimensions using PCA. The 2-D representations of the model parameters, consisting of PCA1 and PCA2, are shown in Fig. 4b. It is evident that while normal data generates mutually-enhanced model parameters, data from devices under

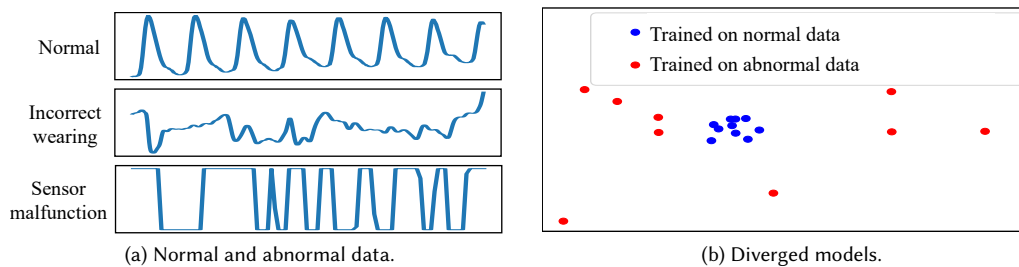


Fig. 4. Impact of abnormal device operations.

abnormal operations leads to a significant divergence in the model parameters of the clients. This divergence negatively impacts the performance of the global model. These observations motivate us to develop an *aggregation* method that recognizes this divergence and can downweight or exclude the contributions of clients with low data quality.

2.4 Systematic Divergence in FRL.

As shown in § 2.1, § 2.2, and § 2.3, divergence within FRL is not a singular issue but rather a multi-level challenge that manifests progressively across three distinct stages: representation, local training, and global aggregation. First, at the foundational representation level, data covariate shift arising from physiological differences among users causes clients to learn misaligned representation spaces. If this fundamental issue is not addressed, client models will build upon inconsistent feature foundations, severely compromising the global model’s generalizability, even with perfect subsequent optimization. This motivates the necessity of a dedicated mechanism at the outset of the learning process to align the representation spaces across all clients. Second, at the local training level, even after achieving representation alignment, the inherent instability of unsupervised learning introduces a new challenge. Without explicit supervisory signals, the optimization trajectory can easily fall into numerous local minima, resulting in a rugged loss landscape. This instability is exacerbated in the FL setting, causing different clients to optimize in divergent directions. Consequently, a strategy is required to stabilize and guide the local optimization process, ensuring all clients proceed along a smooth and consistent path within the aligned representation space. Finally, sensor malfunctions or improper wearing, which lead to the generation of “toxic” local models, can still compromise the optimization stability previously maintained. This compels the design of a final quality control safeguard to dynamically assess and down-weight the contributions of these abnormal clients during aggregation. In summary, divergence in FL is a multi-faceted problem occurring progressively at the representation, local training, and global aggregation levels. Therefore, an integrated framework to address these challenges systematically is required.

3 SYSTEM DESIGN

To address the divergence issues discussed in § 2, we propose Daphne consisting of three key components: i) a client-contrastive learning method that ensures globally consistent representations and addresses covariate shift in decentralized data, ii) an unsupervised curriculum learning strategy that guarantees a stable optimization trajectory during local training, and iii) a learnable weighted aggregation mechanism that mitigates the negative impact of clients with abnormal data at the global aggregation level. The overall design is illustrated in Fig. 5. In the following sections, we provide a detailed explanation of each component’s design.

3.1 Client-contrastive Learning

As discussed in § 2.1, covariate shift in data can lead to misaligned representations among clients, causing divergence among FL clients. The core intuition is that if clients’ feature spaces are misaligned from the outset due to data covariate shifts, any subsequent optimization and aggregation efforts will be severely compromised. Our design rationale is to use contrastive learning to maximize the mutual information between the data representations of different clients, thereby creating a globally consistent representation space. To address this issue, we propose a client-contrastive learning method that enables the model to learn globally consistent representations, as illustrated in Fig. 6a. In the representation space, this method reduces the distance between positive contrastive pairs (sharing the same supervised labels) from the clients and a public dataset, while increasing the distance between negative pairs (having supervised labels) from the

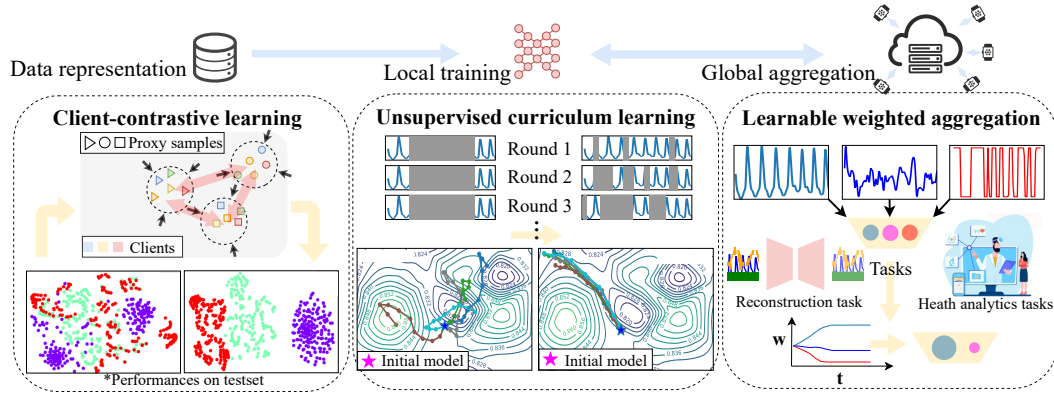


Fig. 5. Overall design of Daphne.

clients and a public dataset. Leveraging this contrastive approach, the clients are compelled to learn globally consistent representations. We detail the operations of the client-contrastive learning method as follows:

Since the data samples are not shared among the clients, Daphne prepares contrastive data samples on the central server. Specifically, the central server maintains a small unlabeled public dataset $\mathcal{D}_s$. In each communication round (from the 2nd round on), the client-contrastive learning method samples N_p instances from $\mathcal{D}_s$ to form a proxy dataset $\mathcal{D}_{\text{proxy}}$. Following this, the encoders from all K client models from the previous round are employed to derive representations for the samples in $\mathcal{D}_{\text{proxy}}$. Each representation $\mathbf{z}_i^k = \text{Encoder}_k(\mathbf{x}_i)$ reflects the output from the k -th client's encoder for the i -th proxy sample. These representations are collectively denoted by $\mathbf{Z}$, where $\mathbf{Z} \in \mathbb{R}^{N_p \times K \times D}$. D , the dimensionality of the representations, is set to 128. It is important to note that since $\mathcal{D}_s$ is a public dataset, sending $\mathcal{D}_{\text{proxy}}$ and $\mathbf{Z}$ to the clients does not pose privacy concerns.

Upon receiving $\mathcal{D}_{\text{proxy}}$ and $\mathbf{Z}$ from the central server, Daphne employs an autoencoder to generate the representations, also aiming to aligning the representations across different clients, thus mitigating the effects of covariate shift. To achieve this, we incorporate a contrastive loss $\mathcal{L}_C$ into the local training loss as follows:

$$\mathcal{L} = \frac{1}{N} \sum_{\mathbf{x} \in \mathcal{D}_{\text{local}}} (f(\mathbf{x}) - \mathbf{x})^2 + \gamma \mathcal{L}_C, \quad (1)$$

where $\mathcal{D}_{\text{local}}$ and N represents the client's local dataset and its size, $f(\cdot)$ denotes the encoding process, and γ is a hyperparameter controlling the weight of $\mathcal{L}_C$. The contrastive loss $\mathcal{L}_C$ is computed utilizing $\mathbf{Z}$ and the $\mathcal{D}_{\text{proxy}}$ provided by the server, along with the representations generated by the current local model. Representations derived from the same proxy sample are considered as positive pairs, whereas those from different samples are treated as negative pairs. In the initial global round, the local loss consists only of the reconstruction loss, given the absence of prior client models to contribute to $\mathbf{Z}$. $\mathcal{L}_C$ is formally expressed as follows:

$$\mathcal{L}_C = -\frac{1}{N_p} \sum_{i=1}^{N_p} \log \frac{\sum_{k=1}^K \langle \mathbf{z}_i^c, \mathbf{z}_i^k \rangle}{\sum_{k=1}^K \langle \mathbf{z}_i^c, \mathbf{z}_i^k \rangle + \sum_{j \neq i}^{N_p} \sum_{k=1}^K \langle \mathbf{z}_i^c, \mathbf{z}_j^k \rangle}, \quad (2)$$

where $\mathbf{z}^c$ are representations generated by the current model's encoder. The similarity $\langle \cdot, \cdot \rangle$ is defined as $\exp(\text{sim}(\cdot, \cdot)/\tau)$, with $\text{sim}(\cdot, \cdot)$ being the cosine similarity between representations, and τ being the temperature scaling parameter. Notably, in the contrastive loss computation, each sample pair must include exactly one sample produced by the current model. Excluding the current model from the contrastive items precludes effective model updates, while incorporating

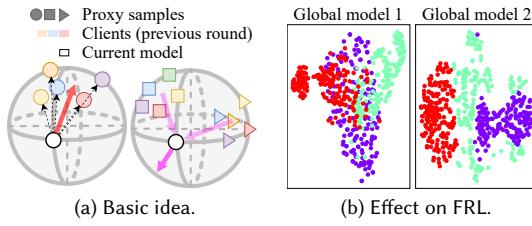


Fig. 6. Client-contrastive learning.

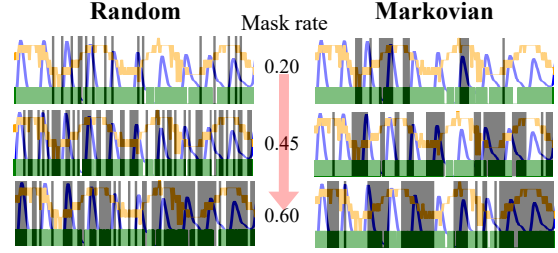


Fig. 7. Increasing Markovian mask rate leads to increased curriculum learning difficulty.

representations from only the current model fails to align the representation spaces across different clients. It is important to clarify that our use of a central server for coordination follows the standard client-server paradigm of FL, where the server’s role is to orchestrate training without accessing raw local user data. Furthermore, the representations shared for contrastive learning are derived from a publicly available proxy dataset, not from users’ private local samples. This process involves an additional transfer of non-sensitive information alongside the standard exchange of model parameters and does not heighten the privacy risks inherent in the standard FL framework.

To validate that client-contrastive learning encourages the model to generate globally consistent representations, we conduct an experiment with 5 clients. Two global models are trained under different FL settings: model 1 without client-contrastive learning, and model 2 with it. Subsequent testing is performed on an unseen subject, with all subjects’ data containing 3 classes. To ensure fairness, the same proxy dataset is provided to both models. Fig. 6b, which employs a t-SNE plot, shows the dimensionally reduced representations. Results indicate that global model 1 struggles to distinguish among the three classes on the subject. In contrast, global model 2 effectively clusters similar samples, forming clear-cut boundaries. This discrepancy arises because model 2 enables clients to learn consistent class representations, whereas covariate shift leads to divergent (hence mixed when aggregated) class representations in model 1. In summary, client-contrastive learning serves as the foundational layer of the Daphne framework. By utilizing a proxy dataset to align representation spaces from the outset, it directly addresses the critical issue that misaligned features caused by data heterogeneity would otherwise severely compromise any subsequent local optimization and global aggregation efforts.

3.2 Unsupervised Curriculum Learning

Building upon the aligned representation space established by client-contrastive learning, we must also ensure that each client’s optimization within this space proceeds consistently toward the global optimum. As discussed in § 2.2, a fixed masking proportion in MAE can destabilize optimization trajectories, causing clients’ optimization directions to diverge. Fortunately, curriculum learning offers a solution by initially presenting simpler tasks and gradually increasing complexity, thereby stabilizing optimization trajectories. While conventional curriculum learning often involves designing a scoring function to assess learning difficulty or integrating difficulty into the loss function, Daphne takes a different approach. It incorporates the concept of increased difficulty of curriculum learning directly into unsupervised training. Specifically, we develop a masking strategy for MAE of Daphne, which generates a variable-length mask for the autoencoder at each global round based on the predefined difficulty level. This strategy not only

avoids the uncertainties and complexities of designing scoring functions but also minimizes the computational overhead associated with calculating scores.

Upon determining a level of difficulty, the most intuitive method for generating masks is to randomly mask each element independently with equal probability. However, this straightforward approach yields discontinuous masks, as illustrated in the left pane of Fig. 7. Unsupervised learning algorithms can then infer the masked values by interpolating from nearby points, thereby significantly undermining the effectiveness of MAE for continuous signals such as physiological data. To overcome this limitation, we propose a Markovian masking strategy. This approach correlates each sample's state with its predecessor, generating a continuous yet random mask. This method compels the MAE to truly learn the underlying signal representations, rather than merely performing interpolation. Specifically, we define a 2×2 state transition matrix $\mathbf{M}$:

$$\mathbf{M} = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix}, p_{00} + p_{01} = p_{10} + p_{11} = 1, \quad (3)$$

where p_{ij} denotes the probability that the current state is i and the next sample state is j . Thus, after obtaining the state S_x (e.g., $[0, 1]$ or $[1, 0]$) of the current sample, the probability distribution for the state of the next sample can be derived as $P_{x+1} = S_x \cdot \mathbf{M}$. In our methodology to augment the difficulty within curriculum learning, we strategically increase the mask size in a progressive manner. Specifically, we incrementally increase p_{01} and p_{11} in each round (both p_{01} and p_{11} should not be greater than 0.95) and set the first sample to be unmasked. Note that the level of difficulty is not determined by an external scoring function but is instead intrinsically linked to and controlled by the mask rate of our MAE: a higher mask rate corresponds to a higher difficulty level, as it requires the model to reconstruct the signal from less information. The curriculum is formed by dynamically advancing the task's difficulty. The right panel of Fig. 7 vividly illustrates that while the Markovian mask adheres to the same masking ratio as the random masking strategy, it engenders a significantly more contiguous mask. This continuity is highly conducive to effectively learning the signals' representations. As we increase the mask rate from 0.2 to 0.6, the difficulty of the curriculum learning correspondingly increases. To summarize, the unsupervised curriculum learning module builds intrinsically upon the aligned representation foundation established in the first stage. Driven by the intuition that a smoother loss landscape fosters consistent optimization trajectories, this strategy ensures that clients not only target the same global optimum but also follow a coherent, stabilized path to reach it.

3.3 Learnable Weighted Aggregation

As mentioned in § 2.3, abnormal device operations can impact data quality, leading to diverged client models. While the preceding modules effectively prevent divergence for normally functioning clients, we still need a crucial safeguard during global aggregation. Real-world scenarios often involve abnormal devices that can contribute erroneous or toxic gradients, undermining the integrity of the global model. Therefore, efforts should be made to prevent low-quality clients' toxic gradient from affecting the convergence of the overall model during aggregation. Based on this observation, we propose a novel learnable weighted aggregation mechanism, as shown in Fig. 8. Unlike traditional aggregation methods, which adjust weights solely based on the quantity of data contributed by each client, our approach incorporates data quality into the weighting process. Specifically, we define $\lambda = [\lambda_1, \lambda_2, \dots, \lambda_K]$ as the weights for clients during aggregation, where all λ values are initially set to $\frac{1}{K}$ and normalized before aggregation $\lambda_i = \frac{\lambda_i}{\|\lambda\|}$. Consequently, the aggregated global model $\mathbf{w}$ becomes: $\mathbf{w} = \sum_{i=1}^K \lambda_i \cdot \mathbf{w}_i$, s.t. $\lambda_i \geq 0, \|\lambda\| = 1$, where $\mathbf{w}_i$ represents the model parameters of the i -th client. The core idea of this module is straightforward: models trained on high-quality data (characterized by

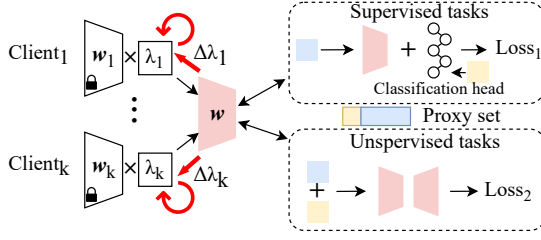


Fig. 8. Process of learnable weighted aggregation.

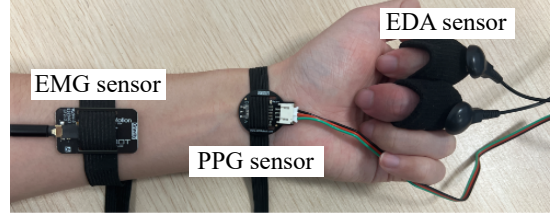


Fig. 9. Hardware implementation of Daphne.

minimal invalid entries and rich information) are expected to perform well on novel tasks. Therefore, if the global model is led by clients with high-quality data, it typically achieves lower losses on designated tasks. Lower-quality clients, due to their contribution to higher losses, receive reduced weights during the optimization process. Consequently, the selection of appropriate multitasks is crucial for this module.

To effectively reflect the data quality of clients participating in aggregation, we configure a multitask framework that incorporates the original unsupervised task along with three health analytics tasks: emotion detection, sleep monitoring and fatigue detection. The selection of these supervised tasks is based on the following considerations: first, all tasks require the involvement of multiple modalities (e.g., PPG, EMG, and EDA), but emphasize different modalities. This choice aims to assess whether the clients' data contain sufficient information pertinent to health analytics from a generalization performance perspective. Second, they encompass tasks of varying complexity, from binary classification to multi-class classification, enabling an examination of data quality across different levels. Additionally, the unsupervised task acts as a baseline, ensuring that the multitask setup does not overfit to any specific health analytics task.

To facilitate client weight optimization within the proposed multitask framework, we continue to use the aforementioned dataset $\mathcal{D}_{\text{proxy}}$, which includes labeled data for three supervised health analytics tasks. It should be noted that the proxy dataset can also be used for the unsupervised learning. For the supervised tasks, we append a single-layer fully connected classification head to the encoder, mapping the representations to the label space. Data for these tasks within $\mathcal{D}_{\text{proxy}}$ are divided into a multitask pre-training set and a multitask training set, with the former used to train the classification head. It is important to clarify the distinction between the data availability on clients and the server. While the label scarcity challenge prevents individual users from annotating their massive private data, the server can reasonably access a small-scale, publicly available dataset to form $\mathcal{D}_{\text{proxy}}$. In practice, system coordinators can easily curate such a proxy set from existing public benchmark datasets. By leveraging this small public dataset to guide the utilization of vast decentralized unlabeled data, Daphne eliminates the unrealistic requirement for large-scale annotation on user devices while ensuring effective model aggregation. To ensure fairness among the tasks, we implement gradient scaling $\nabla \tilde{\mathcal{L}}_T = \frac{\alpha}{\|\nabla \mathcal{L}_T\|} \nabla \mathcal{L}_T$, where T denotes a task within the multitask setup, and $\nabla \tilde{\mathcal{L}}_T$ and $\nabla \mathcal{L}_T$ represent the gradients before and after rescaling, respectively. We select the norm of the unsupervised task as the target norm α . Consequently, during each global round, λ is updated by:

$$\lambda \leftarrow \lambda - \eta \sum_T \nabla_{\lambda} \tilde{\mathcal{L}}_T(\mathcal{D}_{\text{proxy}}, \mathbf{w}), \quad (4)$$

where η is the learning rate. For the unsupervised task (task U), $\mathcal{L}_T$ represents the reconstruction error, specifically the mean squared error (MSE); for the health analytics tasks (task F , S , and E for fatigue detection, sleep monitoring, and

emotion recognition, respectively), $\mathcal{L}_T$ is defined by the cross-entropy loss:

$$\mathcal{L}_T = \begin{cases} -\mathbb{E} [f(\mathbf{x}) - \mathbf{x}]^2 & \text{task } U \\ -\mathbb{E} [\sum_{c=1}^C y_c \log(p_c)] & \text{task } F, S, E, \end{cases}, \quad (5)$$

where y_c denotes the true label for class c (one-hot encoded), and p_c is the predicted probability for class c . It is essential to note that the encoder parameters (as well as the classification head) remain frozen during this process and are not updated.

While this multi-task framework is designed to holistically assess data quality, it also introduces seemingly conflicting optimization objectives. For example, emotion recognition may require sensitivity to signals of physiological arousal, such as volatile, high-frequency patterns, whereas sleep monitoring requires sensitivity to signals of deep rest, which are characterized by stable, low-frequency signals. Furthermore, the unsupervised reconstruction task aims to be information-preserving, while supervised tasks are inherently information-discarding, focusing only on class-discriminative features. However, Daphne’s learnable weighted aggregation transforms these potential conflicting objective into a mechanism for learning a more robust and generalized representation. The optimization of the client weights λ (Equation 4) based on the aggregate loss across all tasks (Equation 5) ensures that the global model favors client updates that contribute to a versatile representation. A model that overfits to a single task or is trained on non-diverse data will incur a high loss on the proxy set, and its influence will be subsequently down-weighted. Consequently, this process acts as a strong regularization mechanism that incentivizes the learning of a multi-faceted representation space that retains sufficient information to perform well across the entire spectrum of health analytics tasks, rather than learning a simple trade-off. Ultimately, the learnable weighted aggregation acts as the final, crucial safeguard in our progressive framework. By functioning as a real-time quality control mechanism that dynamically assesses multi-task performance, it effectively identifies and down-weights “toxic” gradients from abnormal devices, thereby preserving the integrity of the global model.

4 IMPLEMENTATION AND DATASET

In this section, We describe the implementation of and the dataset used by Daphne.

4.1 System Implementation

4.1.1 Hardware. We implement the hardware of Daphne employing multiple sensors to collect multimodal physiological signals, including PPG, EDA, and EMG, as shown in Fig. 9. PPG sensor (DFRobot SKU:SEN0203 [11]) emits green light with wavelengths ranging from 510nm to 540nm. It records changes in light intensity absorbed or reflected by blood, thereby estimating blood volume variations within the vessels. EDA sensor (SICHIRAY EDK0014 [54]) applies a voltage less than 0.5V to the skin and measures the resulting electrical current through the skin to obtain data on skin conductance. This conductance is closely associated with the activity of the autonomic nervous system, particularly reflecting the activity of the sympathetic nervous system. EMG sensor (DFRobot SKU:SEN0240 [12]), placed on the skin, measures weak muscle potentials, typically within a range of ± 1.5 mV. The sampling rates for PPG, EDA, and EMG sensors are set to 64Hz, 64Hz, and 500Hz, respectively.

4.1.2 Software. We utilize 2 NVIDIA GeForce RTX 4090 GPUs for model training. Our software framework is built on Python 3.7, PyTorch 2.1.0, and CUDA 12.1. We update the client data every 20 global rounds to simulate real-world client data dynamics. Detailed configuration is as follows:

- Data from 36 subjects is split into three groups: 30 of them as local data for 30 clients, another 3 for D_{proxy} , and the remaining 3 as the test set.
- Each sample contains 32 seconds of physiological signals. We use 1-d convolution/transpose convolution kernels with a size of 3 in the encoder/decoder.
- In the unsupervised curriculum learning, we initially set p_{01} and p_{11} to 0.2 and 0.6, respectively, and increase them by 0.005 each global round.
- D_{proxy} contains 200 unlabeled samples, sampled from D_s each round, and 600 labeled samples. The contrastive loss weight γ is set to 1.
- We use the Adam optimizer with both the main and multitask learning rate set to 0.001. We also use batch normalization and a 0.3 dropout rate.

4.2 Dataset

The study involves 36 participants aged 25.1 ± 5.5 years, with 18 males and 18 females. Exclusion criteria include pregnancy and mental disorders. We collect two types of data for each participant: unlabeled daily data and labeled task-specific data. For the unlabeled dataset, participants self-collect data for 1.5 hours each in the morning, afternoon, and evening over the course of a week. The labeled dataset involves several tasks. Sleep monitoring requires participants to collect at least 1.5 hours of sleep data daily and rate their sleep quality based on their subsequent work and mental state. Emotion recognition, involves five emotional states: calm (reading neutral magazines), stress (undergoing trier social stress test (TSST) [30]), happy (watching funny videos), angry (engaging in a debate), and sad (listening to sad songs). Fatigue detection involves participants listening to soothing music with eyes closed to reduce fatigue, followed by performing repetitive numeric input tasks to induce fatigue. The labeled datasets, except for the sleep monitoring task, are all collected in the lab. The unlabeled dataset consists of 110,000 samples, with the majority (90%) serving as our main dataset for unsupervised learning and a small portion (10%) from which D_s is sampled for client-contrastive learning. The labeled dataset contains 6,000 samples, primarily (5,400) used as a test set to evaluate the rapid adaptation of our model to new tasks. Crucially, to construct the proxy set D_{proxy} for validating client updates, we reserve the remaining 600 labeled samples, ensuring a strictly balanced distribution with 200 samples for each target task (fatigue detection, sleep monitoring, and emotion recognition). Along with the unlabeled samples sampled from D_s , this balanced composition ensures that D_{proxy} is representative of both the general feature distribution and the specific downstream tasks, minimizing potential bias during aggregation.

5 EVALUATION

In this section, we evaluate Daphne by performing overall, module-by-module, and ablation studies.

5.1 Experiment Setup

To evaluate the performance of Daphne, we compare it with conventional FL algorithm FedAvg [39], and heterogeneous-aware FL algorithms FedProx [34], SCAFFOLD [29], MOON [33], FedU [77] and FRAug [6], with the last two methods specifically designed for FRL. Additionally, we include two centralized training baselines: one trained on the complete dataset and the other trained on a single client’s dataset.

- **FedAvg** averages the parameters from multiple local models to obtain a global model for the next round of local training, without uploading local data.

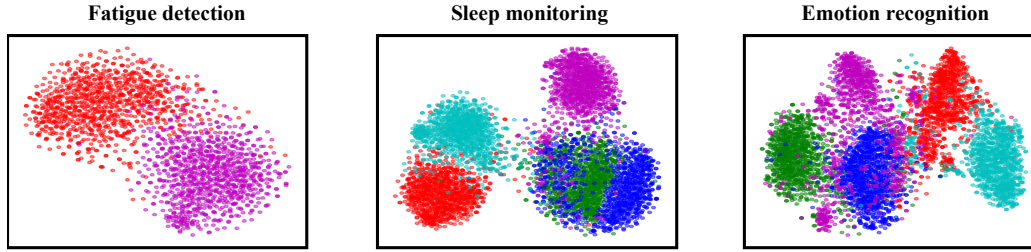


Fig. 10. t-SNE plot for different tasks.

- **FedProx** adds a proximal term in the local loss to reduce the divergence between local and global models, addressing both system and data heterogeneity.
- **SCAFFOLD** uses control variates to align local update directions closer to the global update direction, thereby correcting for client drift.
- **MOON** aligns local models closer to the global model and farther from previous local models through a contrastive loss, improving robustness in heterogeneity.
- **FedU** locally contrasts two Siamese networks and performs conditional aggregation to learn robust representations and mitigate the non-IID problem.
- **FRAug** optimizes a shared embedding generator to augment the client’s training space and mitigate covariate shift.

Note that standard domain generalization (DG) or out-of-distribution (OOD) methods typically require centralized access to pooled source domain data to minimize distribution discrepancies, which violates the strict privacy constraints of federated learning. Furthermore, while a small proxy set exists on the server, it is insufficient in scale to serve as a training source for standard DG methods. Instead, our evaluation includes their federated counterparts that address heterogeneity without raw data sharing: specifically, FedProx acts as a proximal regularizer similar to those used in DG, and MOON specifically employs contrastive OOD learning to handle distribution shifts among clients. Since Daphne is a task-adaptable system based on representation learning, we need to evaluate it on specific health analytics tasks. Specifically, we select fatigue detection, sleep monitoring, and emotion recognition, which are routine tasks that significantly benefit daily life and work, and can be deployed on consumer electronics. Fatigue detection is a binary classification problem, involving two states: alert and fatigued. Sleep monitoring is a 5-class task that assesses sleep quality, with labels 1-5 given by subjects. Emotion detection is a 5-class task involving stressed (ST), happy (HA), calm (CA), angry (AN), and sad (SA). The rating of sleep monitoring is based on a Likert scale and reflects participants’ functional and mental state upon waking, with higher numbers indicating better sleep quality. For each classification task, we select a small portion of the dataset for training a single-layer classification head (few-shot learning), with the remaining data used for testing. The experiments strictly follow the IRB approved by our institution.

5.2 Overall Evaluation of Daphne

We first evaluate the quality of the representations generated by Daphne across different tasks. For each task, we input the data into the encoder, which produces 128-dimensional representations. We then reduce these representations to 2 dimensions using t-SNE for visualization. Note that during this process, labels are only used for the final visualization and not for training any classification head. As shown in Fig. 10, for each task, Daphne clusters data samples of the

same class in the representation space and forms clear-cut boundaries among different classes. This indicates that the representations extracted by Daphne contain sufficient critical information to discriminate various combinations of complex patterns, thereby effectively generalizing to diverse new health analytics tasks.

Based on the representations, we further apply few-shot (15-shot) learning to train a classification head, adapting Daphne for classification tasks. The confusion matrices in Fig. 11 shows the performance of Daphne for different tasks, where samples are highly concentrated along the diagonal. After fine-tuning, Daphne achieves accuracies of 99%, 93%, and 97% on fatigue detection, sleep monitoring, and emotion recognition, respectively. To further illustrate Daphne’s adaptability, we also tested its accuracy with fewer fine-tuning samples (1-shot, 5-shot, 10-shot). The results in Fig. 12 show that even with a single labeled sample (i.e., 30 seconds of physiological signals), an accuracy of 80% can be achieved. The performance improves when the number of labeled samples increases, reaching 90% accuracy with just 10 samples. Daphne’s robust rapid adaptation capability is notably practical, enabling users to accurately tackle new tasks by labeling just a few minutes of data according to the guide displayed on wearable devices (e.g., smartwatches), without the need for complicated operations.

5.3 Superiority of Daphne

We compare Daphne with 6 FL baselines and 2 centralized training baselines. Fig. 13 shows a box plot displaying the accuracy of Daphne and the baselines under 5, 10, and 15-shot fine-tuning. Each box is derived from 50 repetitive experiments, with k samples resampled from the test set for each experiment across all baselines. It is evident that Daphne significantly outperforms all other FL baselines in every scenario. Even in the 5-shot scenario, Daphne achieves an accuracy above 85%, while other FL baselines only reach 40-70%. When the task scenario extends to 15-shot, Daphne achieves a median accuracy of over 93% across three tasks, whereas no other FL baseline exceeds 85%. The superiority of Daphne over the baselines stems from it dealing with divergence at all possible levels in FL. Specifically, Daphne accounts for the divergence in feature distributions among clients caused by covariate shift, which is not addressed by FedAvg, FedProx, SCAFFOLD and FedU. Moreover, although the designs of MOON and FRAug enable them to consider covariate shift, they do not account for the divergence in optimization directions due to unsupervised tasks and the divergence during model aggregation caused by abnormal operations in real-world usage, as Daphne does. An interesting phenomenon is that centralized training (on the entire dataset) also shows lower accuracy than Daphne by 5%-10%. This is because the proposed client-contrastive learning method and unsupervised curriculum learning approach effectively address data and model heterogeneities in FL. Additionally, the learnable weighted aggregation captures and excludes heterogeneity at the global aggregation level, which remains undetected in centralized training.

We further compare the convergence speed of Daphne and the FL baselines. Fig. 14a shows the changes in average loss with increased communication rounds, while Fig. 14b, 14c and 14d illustrate the changes of accuracy in three tasks,

	Fatigue detection		Sleep monitoring					Emotion recognition					
	Alert	Fatigued	1	2	3	4	5	ST	HA	CA	AN	SA	
Alert	.99	.00	1.94	.01	.04	.00	.00	ST	.95	.02	.02	.00	.00
			2.00	.99	.00	.00	.00	HA	.01	.98	.00	.00	.00
			3.01	.00	.87	.02	.07	CA	.01	.01	.97	.00	.00
			4.00	.00	.04	.92	.02	AN	.00	.02	.00	.94	.03
Fatigued	.00	.99	5.00	.00	.06	.02	.91	SA	.00	.00	.00	.00	.98

Fig. 11. Confusion matrices under 15-shot setting.

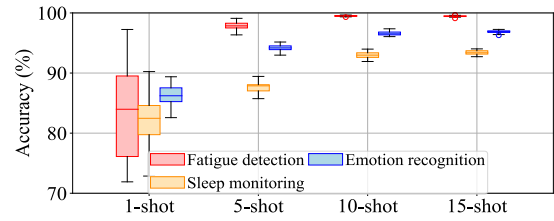


Fig. 12. Accuracies under different few-shot setting.

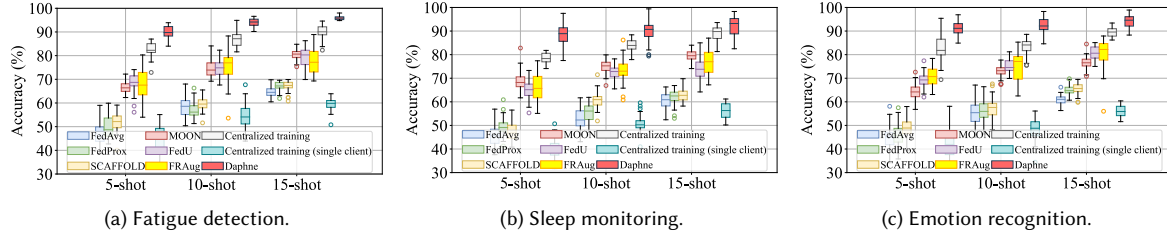


Fig. 13. Comparison with baseline methods.

respectively. The loss is normalized in each baseline and the accuracies are calculated under the 15-shot scenario. It is evident that Daphne exhibits a significantly faster reduction in loss and improvement in accuracy in the initial rounds compared to the baselines. This is because unsupervised curriculum learning effectively prevents high initial learning difficulty from providing unstable optimization directions, thus hindering convergence. Moreover, Daphne achieves convergence within 30 rounds, faster than all baselines, due to the learnable weighted aggregation reducing interference from abnormal clients, allowing Daphne to converge with the assistance of more robust clients. Additionally, Daphne achieves higher accuracy due to the aligned representative space guided by client-contrastive learning.

Finally, We analyze communication overhead incurred in a single communication round, as well as the cumulative overhead until model convergence, as shown in Fig. 15a and 15b. For centralized training, clients transfer all their data to the server at once, making the communication overhead in these two scenarios identical, equal to the total size of all client data. The results indicate that although the per-round communication overhead of Daphne is 25% higher than conventional FL methods that only transmit model parameters, the overall overhead is only 68% of the lowest value of all other methods and just 26% of that of centralized training. The former is due to the client-contrastive learning design, which requires receiving additional $\mathcal{D}_{\text{proxy}}$ and its corresponding representations from the server. However, the latter demonstrates that Daphne’s handling of divergence at three levels reduces convergence time, alleviating concerns about potential communication burden and proving Daphne’s strength in reducing overall overhead.

5.4 Client-contrastive Learning

We first compare the results with and without client-contrastive learning method. Fig. 16a shows the relationship between model accuracy and average client loss in both scenarios, where F , S , and E are fatigue detection, sleep monitoring, and emotion detection, respectively. It is evident that with client-contrastive learning, there is a stronger correlation between client loss and the final model performance. This indicates that optimizing the loss effectively enhances the model’s final performance. Without client-contrastive learning, even if the client loss reaches a relatively low level, the model performance does not improve accordingly. The reason for this difference is whether clients are learning to map the input space to the same representation space. When clients optimize towards a consistent

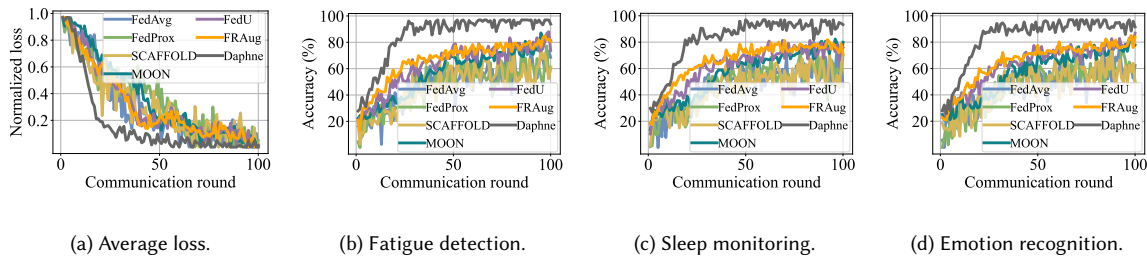


Fig. 14. Comparison during communication rounds.

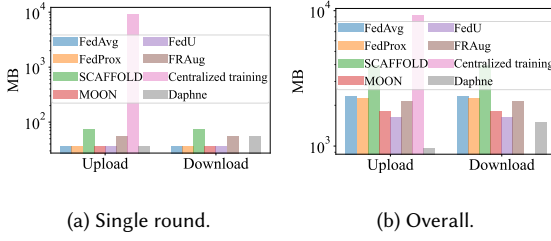


Fig. 15. Communication overhead evaluation.

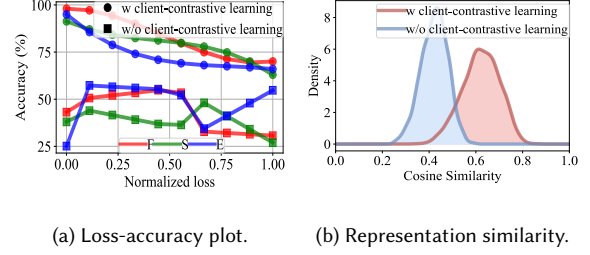


Fig. 16. Impact of client-contrastive learning.

representation space, a decrease in loss indicates that the representation spaces are converging, which mitigates covariate shift and leads to improved overall performance. Conversely, when clients focus solely on their own representation learning, the covariate shift in inputs will be transferred to the representation space. Consequently, even though the local loss decreases, it does not benefit the overall performance.

We further assess the consistency of the features captured by the models through representation similarity. We use the test set to obtain representations from different client models and compute the cosine similarity of the representations obtained for the same sample. Fig. 16b shows the distribution of these similarities. With the constraint of client-contrastive learning, the similarities are concentrated in the range of 0.4-0.8, whereas those without it in the range of 0.2-0.6, with the former being on average at least 0.25 higher than the latter. This indicates that client models under the constraint of client-contrastive learning are more likely to produce consistent representations in new scenarios. Mechanistically, this representation similarity evaluation serves as a variance analysis that proves our contrastive module significantly narrows the variance in representation spaces across clients. Ultimately, this mechanism provides a solid empirical and logical explanation for the system’s enhanced efficacy and generalization capabilities.

5.5 Unsupervised Curriculum Learning

We compare the effectiveness of the unsupervised curriculum learning strategy of Daphne with different mask strategies: fixed mask ratios of 50% and 75%, which are considered to have strong performance, and a curriculum-based strategy that randomly masks each sample with a gradually increasing probability (i.e., the same mask ratio as Daphne in each communication round). Figures 17a shows the trends in average cosine similarity between client gradients as the number of communication rounds increases under different strategies. The result indicates that the gradient directions of MAE with the two fixed mask ratio strategies exhibit significantly larger discrepancies between clients compared to the curriculum-based approach. This is because the fixed mask ratio strategies pose a challenging unsupervised learning task from the start, causing clients to learn towards different latent structures and resulting in non-uniform optimization directions. In contrast, by mitigating the initial learning difficulty, the curriculum-based methods enable clients to focus on a smaller region of feature structures in each round, thus aligning their optimization directions more closely. From a mechanistic perspective, this gradient alignment analysis proves that our curriculum learning strategy effectively forces clients to align their optimization directions much more closely than fixed-mask strategies. This internal consistency acts as the fundamental mechanism for stabilizing local optimization trajectories.

Fig. 17b, 17c, and 17d further demonstrate that Daphne achieves the highest accuracy, outperforming the fixed mask ratio strategies by 17% to 25%, and the curriculum-based random strategy by 28% to 34%. The superior performance

of curriculum-based methods underscores the need to stabilize optimization trajectories among clients. Divergent trajectories result in conflicting gradients during model aggregation, diminishing the efficacy of model updates. Moreover, the Markovian mask employed by Daphne significantly outperforms the independently random mask. This advantage stems from the continuous nature of physiological signals: when the mask is independently sampled for each sample, MAE can easily interpolate the masked value, thereby failing to learn meaningful representations. Daphne mitigates this by generating a continuous mask using a Markov chain.

5.6 Learnable Weighted Aggregation

We evaluate the changes in client weights across communication rounds under different conditions. Our FL setup includes 30 clients, each with an initial weight of 0.033. In the initial state, we set 2 clients engaged in intense activities and 3 clients with sensor malfunctions, while the remaining clients are normal. We replace the data of 2 normal clients with abnormal data at rounds 10 and 13, and replace the data of clients engaged in intense activities to normal at rounds 17 and 20. Fig. 18a demonstrates that Daphne updates the client weights within approximately 5 rounds under various data conditions. It reduces the weights of clients with sensor malfunctions to near 0, while elevating the weights of clients with the best data quality to nearly 0.05. When client data conditions change, there is a 1-2 round delay due to the need for local training. Then, Daphne quickly adapts to these changes, adjusting the client weights accordingly. Experimental results demonstrate that Daphne can continuously evaluate the amount of useful information in client data and quickly detect data quality changes, thereby responding in real-time to potential client divergence. This capability is crucial in real-world scenarios where clients might correct the wearing method or experience loosening during usage. This micro-level dynamic analysis of weight trajectories serves as a direct mechanistic proof of exactly why and how the proposed aggregation works. It demonstrates the framework’s ability to dynamically isolate toxic gradients in real-time, thereby protecting the integrity of the global model.

We then evaluate the effectiveness of the multitask module under five different configurations: including all tasks mentioned in § 3.3; excluding each health analytics task one at a time; and using only the original unsupervised task. Fig. 18b shows the accuracy of Daphne on various test tasks under these configurations, where *F*, *E*, and *S* stand for fatigue detection, emotion recognition, and sleep monitoring, respectively. The configuration that includes all tasks demonstrates the highest performance, achieving nearly 95% accuracy on each test task. Conversely, the configuration with only unsupervised tasks has an accuracy below 90%, as it cannot evaluate the health analytics information in the client data. The configuration with two supervised tasks and one unsupervised task shows slightly lower performance than the full multitask configuration, with accuracy between 90% and 93%. It is noteworthy that for each task, even when excluded from the multitask setting, Daphne still achieves over 90% accuracy. This indicates that the strong performance of Daphne is not due to multitask learning leaking information from the test tasks, but rather its comprehensive assessment of data informativeness.

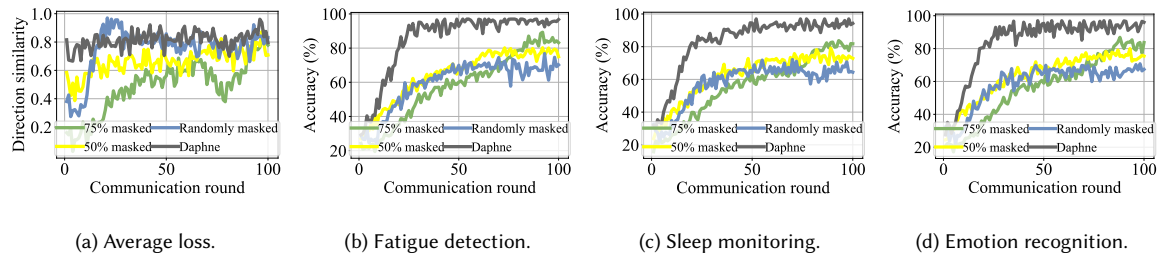


Fig. 17. Comparison of different masking strategies.

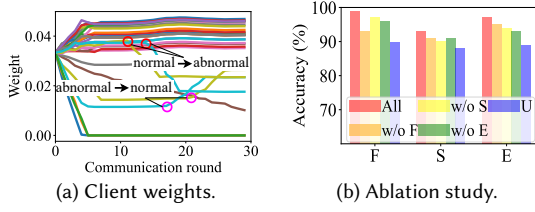


Fig. 18. Learnable weighted aggregation evaluation.

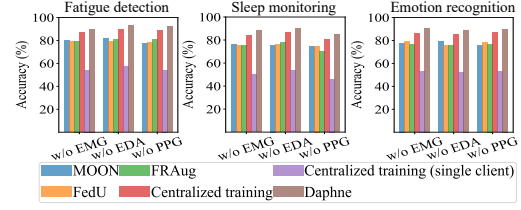


Fig. 19. Ablation study on modalities.

5.7 Ablation Study

We conduct ablation studies at the modality level by removing one modality at a time and training the model with the remaining two modalities. We then evaluate the model’s 15-shot accuracy on three tasks. We compare Daphne against MOON, FedU, FRAug, centralized training with the full dataset, and centralized training with single-client data under these conditions. The results in Fig. 19 show that while Daphne’s performance slightly decreases when some modalities are removed, it maintains at least 85% accuracy across all 9 settings, achieving 90% in 6 of them. Compared to the full-modality scenario, Daphne’s advantage over the FL baselines narrows from 13%-20% to 10%-15%, but it still retains a significant edge. Additionally, it outperforms centralized training with the full dataset by over 3% and maintains a significant 35% advantage over centralized training with single client data.

In these experiments, the reduction in modalities leads to a decrease in information relevant to health analytics, affecting Daphne’s performance. However, Daphne’s robust representation capability allows it to extract useful information from limited modality inputs and maintain excellent performance across various tasks. This has significant practical implications for real-world applications, as sensor configurations in consumer electronics are diverse. Daphne’s representation capability is applicable not only to devices with numerous modalities but also to those with a limited number of modalities.

5.8 Generalization Capability

We assess the generalization capability of Daphne across different users under 15-shot scenario. A leave-one-out experiment is conducted with 20 subjects (out of 36 subjects mentioned in § 4.2) for clarity. Specifically, we train the model on data from 19 subjects and test it on the remaining one, with results displayed in Fig. 20. For all the 20 subjects, Daphne achieves at least 92% accuracy across all tasks, with several reaching 95%, indicating that Daphne’s performance is robust to variations in user age and gender.

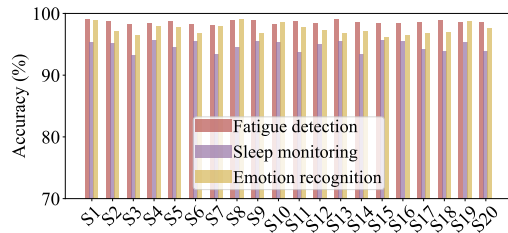


Fig. 20. Performance on different users.

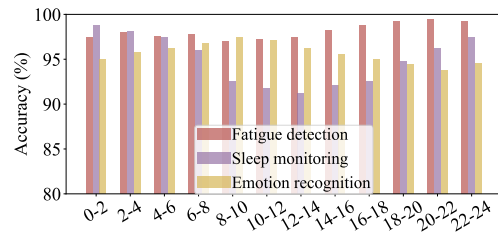


Fig. 21. Performance on different times of the day.

We further evaluate Daphne’s performance throughout different times of the day in a 15-shot scenario. As shown in Fig. 21, although Daphne’s performance exhibits slight fluctuations, it maintains an accuracy above 90% across all tasks, with fatigue detection exceeding 95%. This is because human physiological indicators exhibit periodic changes throughout the day. For instance, emotions tend to be more unstable late at night or early in the morning, affecting emotion recognition. Similarly, sleep patterns differ between daytime and nighttime, impacting sleep monitoring performance. Nonetheless, Daphne demonstrates strong overall performance, indicating that it learns a robust representation space that effectively mitigates such distribution shift in health analytics tasks.

To quantitatively evaluate how representations are influenced by participant characteristics, we conduct an experiment comparing the representations learned by a original FL baseline versus Daphne, focusing on participant age and height. For the age experiment, we partition participants into 2-year interval slots (19 to 31 years old). We then compute the average cosine similarity between representations of sample pairs that share the same label (e.g., “happy”) but are drawn from different age slots. A similar methodology is applied to the height experiment, with participants grouped into 5 cm intervals (155 cm to 185 cm). The results are summarized in Table 1. For the original FL baseline, the representations are heavily influenced by participant characteristics, showing low cosine similarity for both age and height variations. This indicates that the baseline model tends to confuse user-specific variations with the core task, hindering its ability to learn a generalized representation. In contrast, Daphne exhibits a more selective behavior. For age, a feature highly correlated with health indicators, the representation similarity remains lower (0.60-0.73), suggesting the model correctly preserves these meaningful physiological variations. Conversely, for the less correlated feature of height, the similarity is significantly higher (0.84-0.98), indicating the model successfully learns to be agnostic to this irrelevant variation. This comparative analysis demonstrates that, unlike standard FL, Daphne learns a selective, task-aware variation-agnostic representation, distinguishing between meaningful signals and user-specific noise. Fundamentally, this cross-user generalization is driven by the client-contrastive learning module, which forces heterogeneous clients to align with a common proxy space, effectively decoupling and stripping away individual structural covariate shifts.

Table 1. Cosine similarity of representations across age and height variations for different tasks.

Method	Variation	Sleep monitoring	Emotion recognition	Fatigue detection
Daphne	Age	0.62	0.73	0.60
	Height	0.94	0.84	0.98
Original FL	Age	0.31	0.40	0.28
	Height	0.56	0.52	0.49

5.9 Hyper-parameter Search

5.9.1 Unsupervised Curriculum Learning Parameters. In § 3.2, the state transition matrix $\mathbf{M}$ of the Markov chain has two critical parameters p_{01} and p_{11} . These parameters represent the probabilities of the next sample being masked, given the current sample is unmasked or masked, respectively. For optimal mask continuity, we set p_{01} and p_{11} to fall in the ranges of $[0.1, 0.4]$ and $[0.6, 0.9]$, respectively. Fig. 22a presents heatmaps of model accuracies across three tasks, highlighting that $p_{01} = 0.2$ and $p_{11} = 0.6$ are optimal. These values provide a moderate initial difficulty level, fostering better model performance. The heatmaps reveal that low initial difficulty (lower-left corners) results in lower accuracies

because the ease of the reconstruction task fails to promote deeper representation learning. Conversely, high initial difficulty (upper-right corners) leads to optimization divergence, rendering them suboptimal.

As an important parameter in curriculum learning, the difficulty growth rate β increases p_{01} and p_{11} at each communication round. Considering that communication rounds typically range between 10^1 and 10^2 , setting β too high is impractical. We explored optimal values within the interval $[0.001, 0.01]$. Fig. 22b, which includes a line graph, illustrates the model’s accuracy across various β values. When β is too low, the accuracies of the three tasks range from 89% to 94%, which is lower than the range of 93% to 99% observed when β is set to 0.005. When β is increased to 0.1, the accuracies sharply drop to around 82%. This occurs because a low β causes the task difficulty to increase slowly, making it challenging to learn effective representations from simple tasks, while a high β leads to instability in local training.

5.9.2 Client-contrastive Learning Parameter. As mentioned in § 3.1, in the local training loss, γ determines the participation extent of contrastive learning. We perform a search within $[0.5, 3]$, with the results shown in Fig. 22c. It can be observed that the overall accuracies of 3 tasks initially increase and then decrease as γ increases. When γ is 0.5, none of the accuracies is higher than 82% as the contrastive loss component is too small to mitigate the divergence issue. When γ is set to 1, the accuracies are maximized, all exceeding 92%. However, when γ is set to 3, the accuracies drop to under 90% due to overfitting to the proxy dataset $\mathcal{D}_{\text{proxy}}$.

The temperature scaling parameter τ in the contrastive loss determines the strength of contrastive learning. As shown in Fig. 22d, the model performs best when τ is set to 1, achieving accuracies higher than 93%. When τ is set to 0.1, the model becomes unstable due to excessively strong contrastive learning. Conversely, when τ is set to a very large value like 10000, the accuracies drop to 68% to 74% because the contrastive loss component degenerates into a constant and becomes ineffective.

5.9.3 Sensitivity to Proxy Set Configuration. The proxy set $\mathcal{D}_{\text{proxy}}$ serves as the ground truth for measuring client utility. We evaluate Daphne’s sensitivity to both the size and distribution balance of this set. We first randomly downsampled $\mathcal{D}_{\text{proxy}}$ from 100% to 75% and 50% while maintaining class balance. The result shows that the system performance remains robust. Specifically, under the 15-shot setting, reducing the proxy set size by 25% and 50% results in only a 2.3% and 5.2% drop in the average accuracy across all three tasks. This indicates that Daphne relies on the representativeness rather than the absolute volume of proxy data.

To simulate proxy set bias, we then introduced task-level imbalance by reducing the sample count of specific tasks while keeping others fixed. We define the imbalance ratio (ρ) as the ratio of the sample count of the majority task to the minority task. We tested scenarios with $\rho = 2$ and $\rho = 5$ (e.g., reducing emotion recognition samples to 50% and 20% of other tasks). The results show that even under significant imbalance ($\rho = 5$), the global model’s accuracy

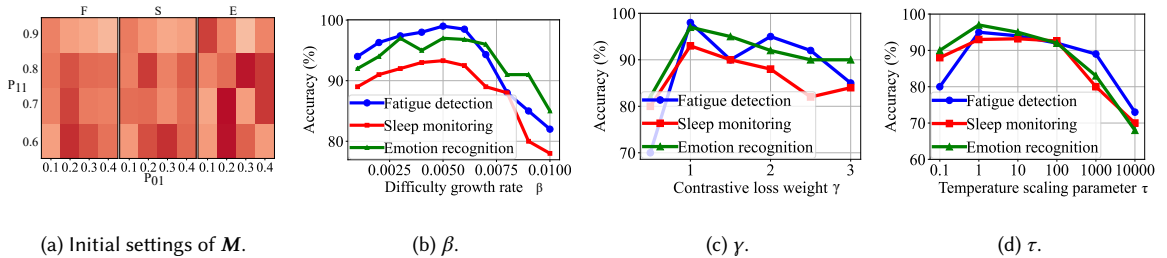


Fig. 22. Hyper-parameter search.

fluctuation on the minority tasks remains within 4.4%. This suggests that as long as the proxy set maintains minimal coverage for each task, the learnable weighted aggregation can effectively filter toxic gradients. These empirical results demonstrate that a strictly balanced proxy dataset is not required, thereby bridging our experimental findings with real-world applicability and making our assumption highly realistic for deployment.

6 CONCLUSION

Taking an important step towards intelligent and pervasive health monitoring, we propose Daphne in this paper for federated task-adaptable health analytics. By employing divergence-dealing strategies at the representation, local training, and global aggregation levels, the Daphne framework manages the vast amounts of underutilized data collected by wearable devices. This approach addresses various divergence issues in the learning pipeline, thereby unlocking the full potential of this data for diverse health-related tasks. Through extensive experiments conducted under highly heterogeneous scenarios and comparisons with other baselines, we have demonstrated the promising performance of Daphne in task-adaptable health analytics. We plan to further scale our experiments to promote the real-life usage and broader acceptance of the Daphne framework. Furthermore, while extensive empirical and mechanistic analyses demonstrate the robustness of Daphne, we acknowledge the current challenge of establishing formal mathematical convergence guarantees. Specifically, real-world multimodal physiological signals are highly non-stationary and exhibit complex, non-IID distributions. Establishing rigorous mathematical bounds for such systems typically requires strong, simplifying assumptions that are fundamentally disconnected from the reality of health analytics. Consequently, deriving tight and meaningful theoretical proofs in this specific context is currently intractable, yet exploring these theoretical boundaries remains a critical direction for our future work.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China under grant No. 62502191, the Guangdong Basic and Applied Basic Research Foundation under grant No. 2026A1515011651, and Shenzhen Science and Technology Program under grant No. 20231120215201001.

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